Scribble Segmentation & ShapePU

Yuanye Liu

yuanyeliu7@gmail.com

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Outline

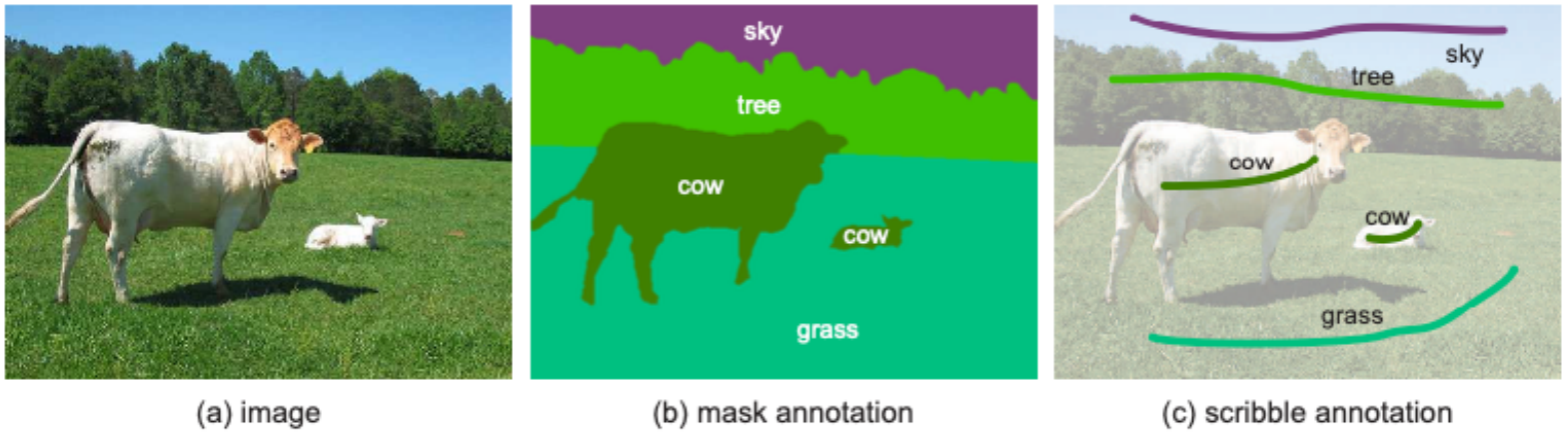
1 Scribble Segmentation

2 ShapePU

Introduction

Scribble annotations

1. Easily accessible : The scribble annotations are common and easy to obtain.
2. Widely used : The scribble annotations are widely used in academic research and commercial software.
3. User-friendly : User-friendly way of interacting.



(a) image (b) mask annotation (c) scribble annotation

Figure: Lin, Di, et al. "Scribblesup: Scribble-supervised convolutional networks for semantic segmentation." 2016.

Introduction

Scribble vs. Image-level vs. Box-level

1. Image-level : scribbles provide location information at a few pixels, which should lead to better results.
2. Box-level : scribbles are lack determinate bounds of objects and more ambiguous.

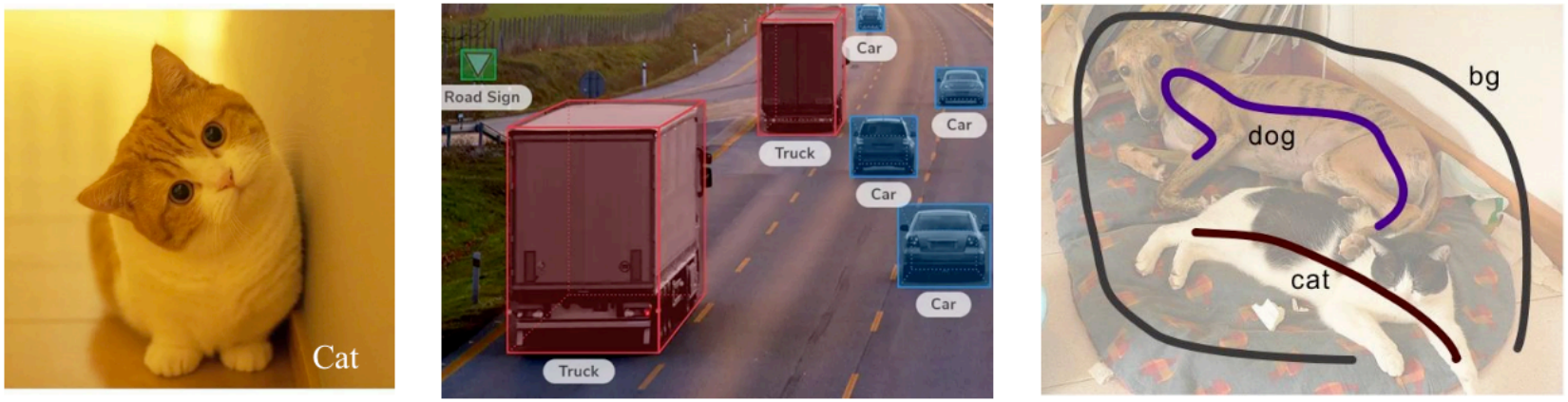


Figure: Image-level vs. Box-level vs. Scribble

Scribblesup: Scribble-supervised convolutional networks for semantic segmentation

Two tasks: dependent on each other

1. Propagate the semantic labels from scribbles to other pixels and fully annotates the images.
2. Learn a convolutional network(FCN) for semantic segmentation.

Figure: Two tasks

Scribblesup: Scribble-supervised convolutional networks for semantic segmentation

Objective function

$$\sum_i \psi_i \left(y_i \mid X, S\right) + \sum_{i,j} \psi_{ij} \left(y_i, y_j \mid X\right)$$

1. X: training image; x_i: super-pixels.($\bigcup_i x_i = X$)
2. $S = \{s_k, c_k\}$. s_k:pixels of scibble k; c_k:category label of scribble k.
3. $Y = \{y_i\}$: label of pixel x_i
4. ψ_i: unary term; ψ_{ij}: pairwise term.

Objective function

Unary term: $\psi_i = \psi_i^{scr} + \psi_i^{net}$

$$\psi_i^{scr}(y_i) = \begin{cases} 0 & \text{if } y_i = c_k \text{ and } x_i \bigcap s_k \neq \varnothing \\ -\log\left(\frac{1}{|\{c_k\}|}\right) & \text{if } y_i \in \{c_k\} \text{ and } x_i \bigcap S = \varnothing \\ \infty & \text{otherwise} \end{cases}$$

$$\psi_i^{net}(y_i) = -\log P(y_i \mid X, \Theta)$$

Objective function

Pairwise term

$$\psi_{ij}(y_i, y_j \mid X) = [y_i \neq y_j] \exp\left\{ -\frac{\|h_c(x_i) - h_c(x_j)\|_2^2}{\delta_c^2} - \frac{\|h_t(x_i) - h_t(x_j)\|_2^2}{\delta_t^2} \right\}$$

1. h_c: color histogram, RGB space, 25bins.
2. h_t: texture histogram, horizental and vertical orientations, 10bins.
3. δ_c, δ_t: constant, *e.g.* $5, 10$.

Objective function

$$\sum_i \psi_i^{scr}(y_i \mid X, S) + \sum_i -\log P(y_i \mid X, \Theta) + \sum_{i,j} \psi_{ij}(y_i, y_j \mid X)$$

Optimization

Alternative process

1. Fix Θ solve for Y: the solver propagates labels to unmarked pixels, based on scribbles, apperance, and also network predictions.
2. Fix Y solve for Θ: the solver learns a fully-convolutional network for pixel-wise semantic segmentation.

Optimization

Fix Θ solve for Y

Propagating scribble information to unmarked pixels

1. Unary term ψ_i: evaluated by enumerating all possible labels.
2. Pairwise term ψ_{ij}: pre-computed as a look-up table.
3. Solve objective function by multi-label graph cuts solution.

Fix Y solve for Θ

Optimizing network parameters

1. With labeling Y fixed, optimize the network(FCN) parameters Θ using Y as supervision.

Positive Unlabeled Learning

Introduction

1. Binary Classification: need fully annotated on positive and negative samples.
2. PU learning:Only some of positive examples are labeled.
3. One reason that PU learning has attracted attention is that PU data naturally arises in many significant applications.
   - i Personalized advertisement
   - ii Medical records
   - iii Knowledge base completion

ShapePU

Network

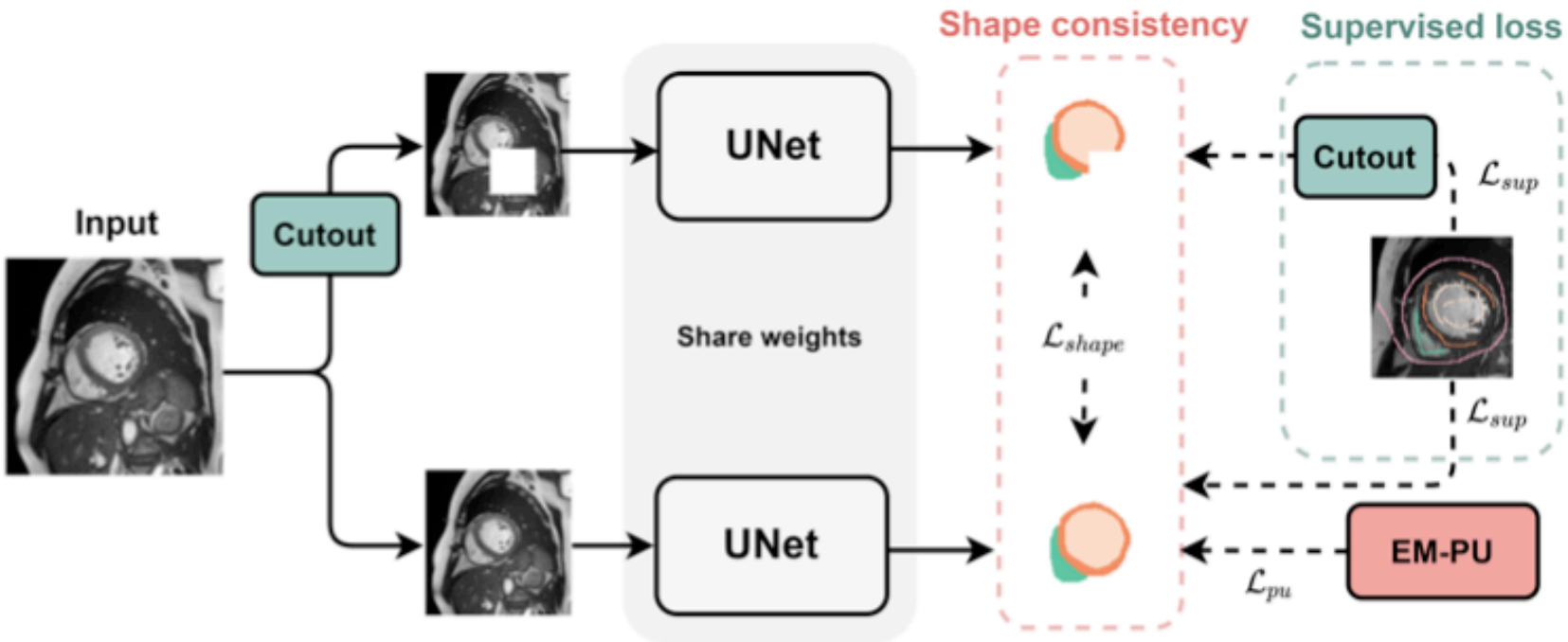


Figure: Zhang K, Zhuang X. ShapePU: A New PU Learning Framework Regularized by Global Consistency for Scribble Supervised Cardiac Segmentation[J]2022.

ShapePU

EM Estimation of Multi-class Mixture Proportion

1. $P_u = \sum_{j=0}^{m} \alpha_j P_j$; P_u: unlabeled pixels distribution, $\{P_i\}$: all label distribution. α_j: mixture proportion.
2. Assumption: $\hat{p}_l\left(x \mid c_j\right) = \frac{p_l(c_j|x)p(x)}{p_l(c_j)} = \frac{p_u(c_j|x)p(x)}{p_l(c_j)} = \hat{p}_u\left(x \mid c_j\right)$
3. $\alpha_j = \frac{1}{n_u} \sum_{i=1}^{n_u} \hat{p}_u\left(c_j \mid x_i\right) = \frac{1}{n_u} \sum_{i=1}^{n_u} \frac{\hat{p}_u(c_j)\hat{p}_l(c_j|x_i)/\hat{p}_l(c_j)}{\sum_{j=0}^{m}[\hat{p}_u(c_j)\hat{p}_l(c_j|x_i)/\hat{p}_l(c_j)]}$

ShapePU

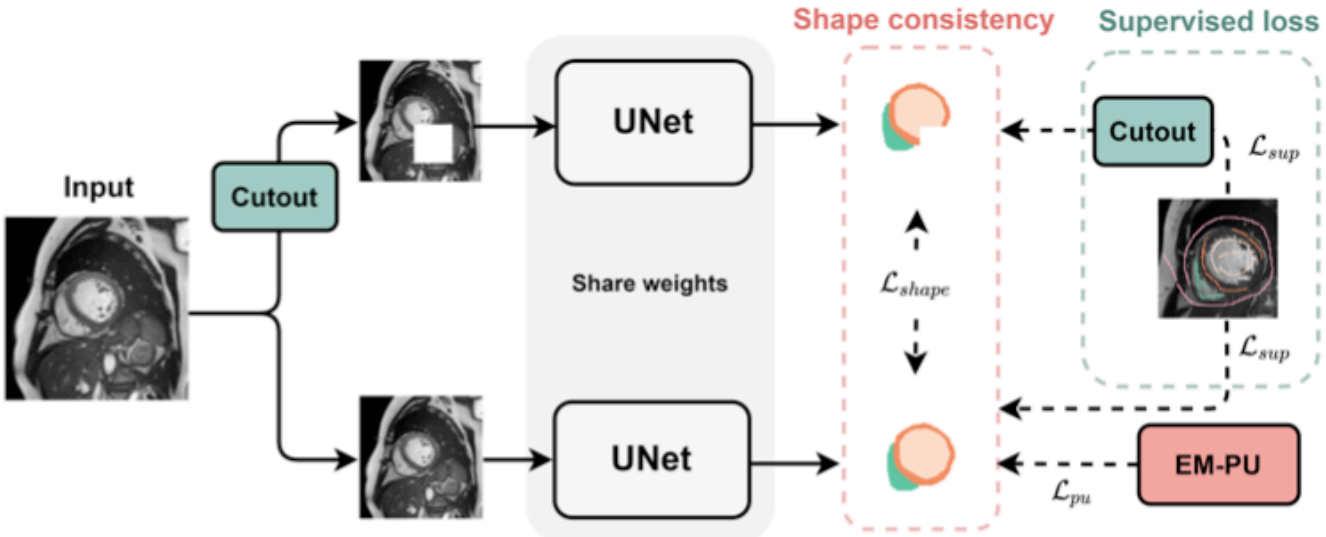
PU Learning with Marginal Probability

1. Rank the unlabeled pixels according to their probability belongs to class c_j. The pixels within the α_j proportion are considered as positive samples.
2. Apply a loss function to the set of predicted negative samples $\bar{\Omega}_j$.
3. $\mathcal{L}^- = -\sum_{j=1}^{m} \sum_{i \in \bar{\Omega}_j} \log\left(\hat{p}\left(\bar{c}_j \mid x_i\right)\right)$

ShapePU

Global Consistency

1. $X, X', f(X), f(X'), f'(X)$
2. $\mathcal{L}_{\text{global}} = \frac{1}{2}\mathcal{L}_{\cos}\left(f'(X), f(X') + \frac{1}{2}\mathcal{L}_{\cos}\left(f(X'), f'(X)\right)\right.$



ShapePU

Loss function

1. $\mathcal{L}^{+} = -\sum_{i \in \Omega_l} \sum_{j=1}^{m} \left[y_{ij} \log \left(\hat{y}_{ij} \right) \right]$
2. $\mathcal{L} = \mathcal{L}^{+} + \lambda_1 \mathcal{L}^{-} + \lambda_2 \mathcal{L}_{\text{global}}$

Thank you
for
listening!